

Critical Path Analysis — Pier P4 Ground Condition and Plant Breakdown

Nandira River Bridge Project — Package NRB-4

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Contract No.: NHIA/NRB4/CW/2020-01
From: Planning Engineer, Sagara Constructions Pvt. Ltd.
To: Engineer, Meridian Engineering Consultants

Subject: Critical Path Analysis Submitted in Response to ENG-NRB4-0065 — Pier P4 Pile Position P4-N4

In response to your letter ENG-NRB4-0065 dated 10-Jan-2023, we submit the following critical path analysis isolating the independent effect of the ground condition and the plant breakdown on progress at pile position P4-N4.

PERIODS ANALYSED:

Cause	Start	End	Independent Duration
Ground condition (P4-N4)	03-Nov-2022	15-Dec-2022	6 weeks
Plant breakdown (Rig PR-2)	07-Nov-2022	28-Nov-2022	3 weeks
Overlap (both causes operative)	07-Nov-2022	28-Nov-2022	3 weeks

1. GROUND CONDITION IN ISOLATION: Had Rig PR-2 remained available throughout, boring at pile position P4-N4 could not have proceeded to the revised design before completion of Geotechnical Investigation Report GEO-NRB4-004 (16-Nov-2022) and Site Instruction SI-NRB4-027 (22-Nov-2022), and, allowing for the additional 3.5m socket depth instructed, would still have required a total period of 6 weeks from 03-Nov-2022 to 15-Dec-2022 to complete.
2. PLANT BREAKDOWN IN ISOLATION: Had the ground condition not been encountered, boring at pile position P4-N4 to the original design would have continued using Rig PR-2 until its breakdown on 07-Nov-2022, and would have resumed on the rig's return to service on 28-Nov-2022, a standalone impact of 3 weeks.
3. COMBINED EFFECT: The 3-week period during which Rig PR-2 was out of service (07-Nov-2022 to 28-Nov-2022) falls entirely within the 6-week ground condition period. The critical path to completion of pile position P4-N4 was, throughout this 6-week period, controlled by the ground condition and the redesign process it necessitated: the geotechnical investigation and Site Instruction SI-NRB4-027 were still outstanding for the greater part of the period Rig PR-2 was unavailable, such that the unavailability of the rig did not, in our assessment, independently extend the date pile position P4-N4 was actually completed.
- We maintain, on this basis, that the ground condition is the controlling cause of the full 6-week delay to pile position P4-N4, and that our claim for a 6-week extension under CTR-NRB4-0142 stands. We confirm the dates and durations in this analysis are drawn from the contemporaneous records already provided, including

EQR-NRB4-001, GEO-NRB4-004 and SI-NRB4-027.

Prepared by: Planning Engineer, Sagara Constructions Pvt. Ltd.